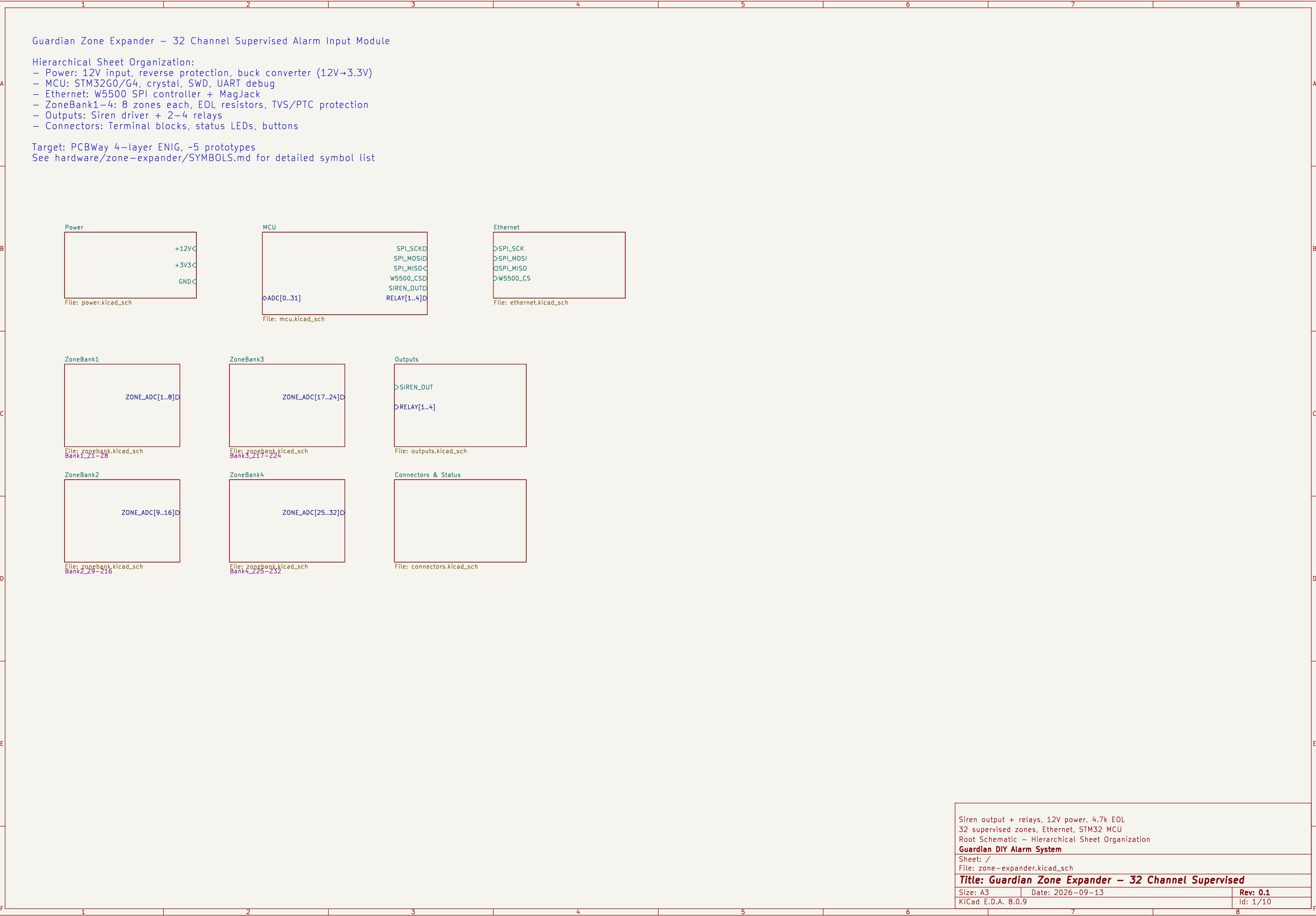
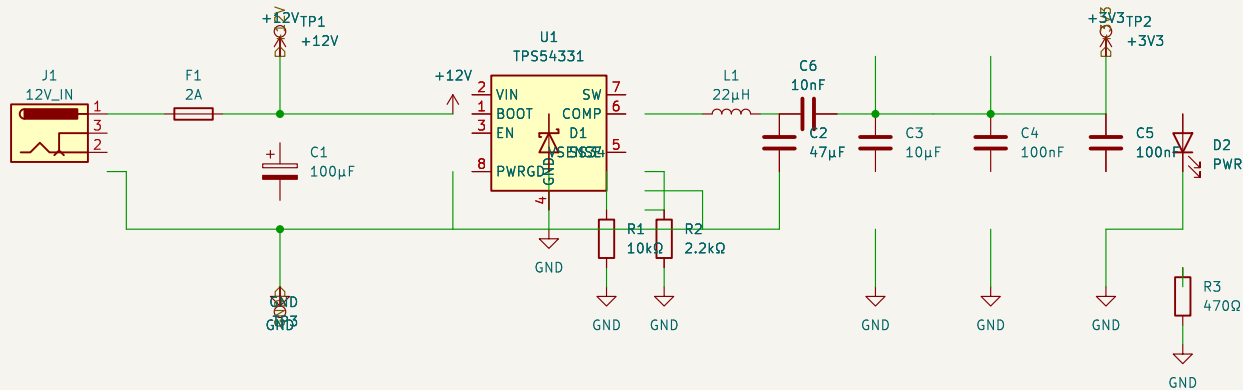






Power Input: 12V DC barrel jack, 2A fuse
 Buck converter: TPS54331 (12V → 3.3V @ 1.5A)
 Output ripple < 50mV pk-pk
 Output current: 1A typical, 1.5A max

Input fusing, decoupling, test points
 Buck converter: 12V → 3.3V @ 1.5A
 12V DC input with reverse protection
Guardian DIY Alarm System

Sheet: /Power/
 File: power.kicad_sch

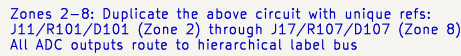
Title: Guardian Zone Expander – Power Supply

Size: A4 Date: 2026-09-13

KiCad E.D.A. 8.0.9

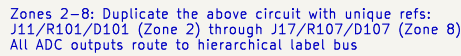
Rev: 0.1

Id: 2/10



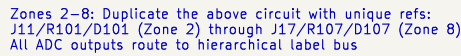
This bank implements 8 identical channels.
Replicate this sheet 4 times for 32 total zones.

TVS protection, pull-up resistors, ADC inputs		
EOL 4.7kΩ supervision: detect normal/alarm/open/short		
Reusable 8-channel zone input bank		
Guardian DIY Alarm System		
Sheet: /ZoneBank1/		
File: zonebank.kicad_sch		
Title: Guardian Zone Expander – Zone Bank (8 Channels)		
Size: A3	Date: 2026-09-13	Rev: 0.1
KiCad E.D.A. 8.0.9		Id: 3/10



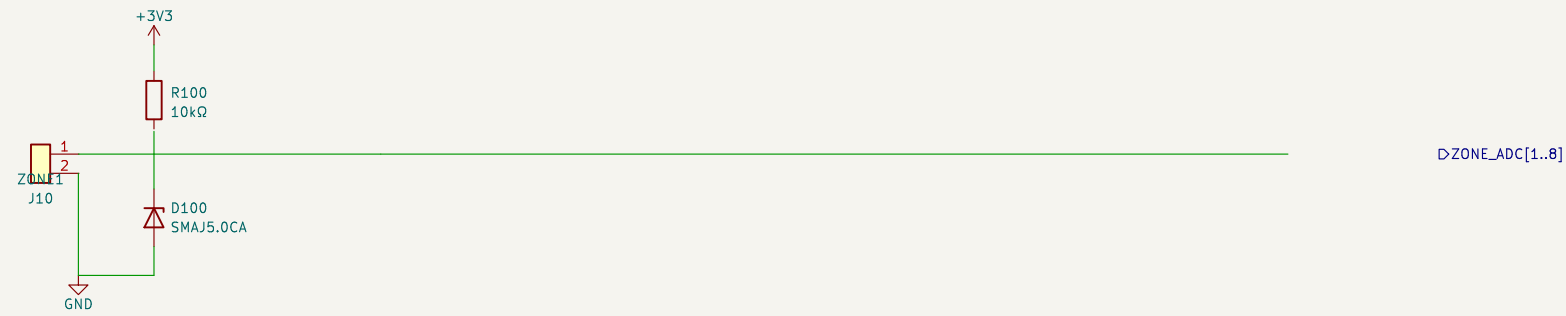
This bank implements 8 identical channels.
Replicate this sheet 4 times for 32 total zones.

TVS protection, pull-up resistors, ADC inputs		
EOL 4.7kΩ supervision: detect normal/alarm/open/short		
Reusable 8-channel zone input bank		
Guardian DIY Alarm System		
Sheet: /ZoneBank2/		
File: zonebank.kicad_sch		
Title: Guardian Zone Expander – Zone Bank (8 Channels)		
Size: A3	Date: 2026-09-13	Rev: 0.1
KiCad E.D.A. 8.0.9		Id: 4/10



This bank implements 8 identical channels.
Replicate this sheet 4 times for 32 total zones.

TVS protection, pull-up resistors, ADC inputs		
EOL 4.7kΩ supervision: detect normal/alarm/open/short		
Reusable 8-channel zone input bank		
Guardian DIY Alarm System		
Sheet: /ZoneBank3/		
File: zonebank.kicad_sch		
Title: Guardian Zone Expander – Zone Bank (8 Channels)		
Size: A3	Date: 2026-09-13	Rev: 0.1
KiCad E.D.A. 8.0.9		Id: 5/10



Zones 2-8: Duplicate the above circuit with unique refs:
J11/R101/D101 (Zone 2) through J17/R107/D107 (Zone 8)
All ADC outputs route to hierarchical label bus

Zone Front-End Design (per channel):

ZONE+ terminal → 10kΩ pull-up to +3.3V → ADC input
↓
TVS diode to GND (5.6V bidirectional)
↓
ZONE- terminal (GND)

Sensor loop: ZONE+ → contact → 4.7kΩ EOL resistor → ZONE-

ADC voltage states:

- Normal (loop closed, 4.7kΩ): ~1.55V (divider: $4.7k\Omega / (10k\Omega + 4.7k\Omega) \times 3.3V$)
- Alarm (loop short, 0Ω): ~0V
- Open/Tamper (loop open, ∞Ω): ~3.3V (full pull-up)
- Fault: Intermediate voltage indicates wrong/damaged EOL

This bank implements 8 identical channels.
Replicate this sheet 4 times for 32 total zones.

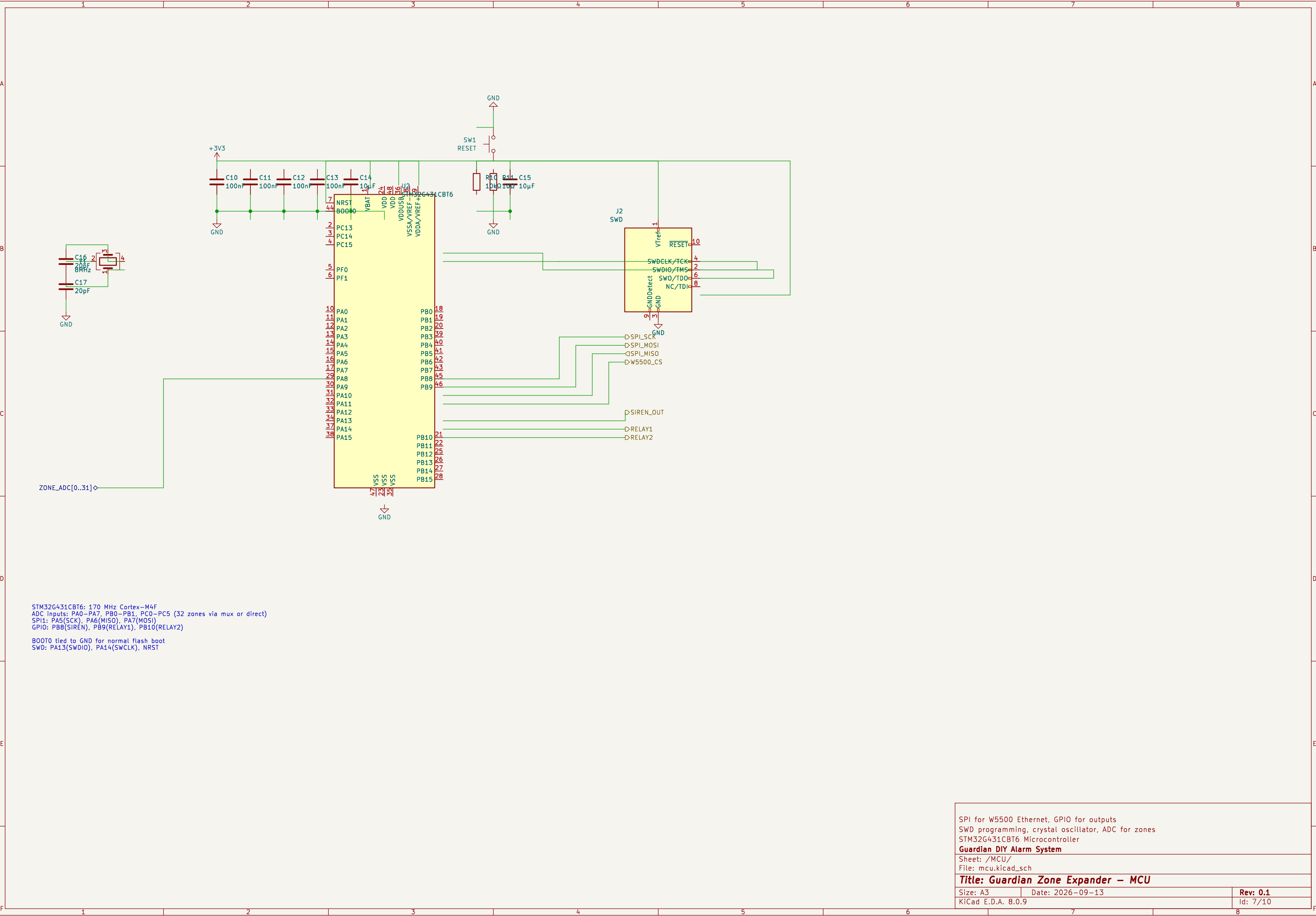
- TVS protection, pull-up resistors, ADC inputs
- EOL 4.7kΩ supervision: detect normal/alarm/open/short
- Reusable 8-channel zone input bank

Guardian DIY Alarm System

Sheet: /ZoneBank4/
File: zonebank.kicad_sch

Title: Guardian Zone Expander – Zone Bank (8 Channels)

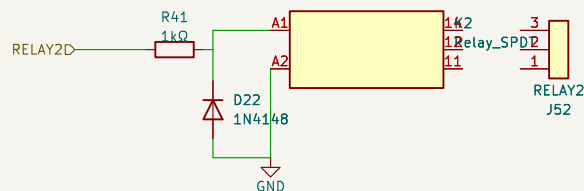
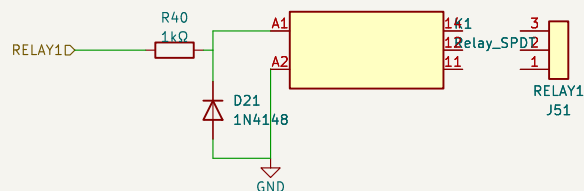
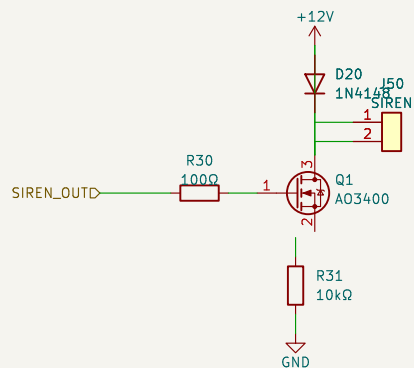
Size: A3	Date: 2026-09-13	Rev: 0.1
KiCad E.D.A. 8.0.9		Id: 6/10



STM32G431CBT6: 170 MHz Cortex-M4F
ADC inputs: PA0-PA7, PB0-PB1, PC0-PC5 (32 zones via mux or direct)
SPI1: PA5(SCK), PA6(MISO), PA7(MOSI)
GPIO: PB8(SIREN), PB9(RELAY1), PB10(RELAY2)

BOOT0 tied to GND for normal flash boot
SWD: PA13(SWDIO), PA14(SWCLK), NRST

SPI for W5500 Ethernet, GPIO for outputs SWD programming, crystal oscillator, ADC for zones STM32G431CBT6 Microcontroller Guardian DIY Alarm System		
Sheet: /MCU/ File: mcu.kicad_sch		
Title: Guardian Zone Expander – MCU		
Size: A3	Date: 2026-09-13	Rev: 0.1
KiCad E.D.A. 8.0.9		Id: 7/10



Siren: N-channel MOSFET (logic-level) drives 12V siren load
Flyback diode protects MOSFET from inductive kick

Relays: 5V or 12V coil, SPDT (Form C) for dry contact switching
Flyback diodes on coils
Terminal blocks for NO/COM/NC outputs

Relay drivers with flyback diodes
MOSFET driver for 12V siren with flyback protection
Siren driver + 2× Form-C relays

Guardian DIY Alarm System

Sheet: /Outputs/
File: outputs.kicad_sch

Title: Guardian Zone Expander – Outputs

Size: A4	Date: 2026-09-13	Rev: 0.1
KiCad E.D.A. 8.0.9		Id: 8/10



Mounting holes: 4x M3 for DIN rail or enclosure

Power / Link / Status indicators Status LEDs, mounting holes Guardian DIY Alarm System		
Sheet: /Connectors & Status/ File: connectors.kicad_sch		
Title: Guardian Zone Expander – Connectors & Status		
Size: A4	Date: 2026-09-13	Rev: 0.1
KiCad E.D.A. 8.0.9		Id: 9/10

